Receipt of Public and Private Food Assistance Across the Rural-Urban Continuum Before and During the COVID-19 Pandemic: Analysis of Current Population Survey Data

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Abstract

Background: The nutrition safety net in the United States is critical to supporting food security among households in need. Food assistance in the United States includes both government-funded food programs and private community-based providers who distribute food to in need households. The COVID-19 pandemic impacted experiences of food security and use of private and public food assistance resources. However, this may have differed for households residing in urban versus rural areas. We explored receipt of Supplemental Nutrition Assistance Program (SNAP) benefits or food from community-based emergency food providers across a detailed measure of the rural-urban continuum before and during the COVID-19 pandemic.

Methods: We linked restricted use Current Population Survey Food Security Supplement data to census-tract level United States Department of Agriculture Rural-Urban Commuting Area codes to estimate prevalence of self-reported SNAP participation and receipt of emergency food support across temporal (2015-2019 versus 2020-2021) and socio-spatial (urban, large rural city/town, small rural town, or isolated rural town/area) dimensions. We report prevalences as point estimates with 95% confidence intervals, all weighted for national representation.

Results:

The weighted prevalence of self-reported SNAP participation was 8.9% (8.7-9.2%) in 2015-2019 and 9.1% (8.5-9.5%) in 2020-2021 in urban areas, 11.4% (10.8-12.2%) in 2015-2019 and 11.6% (10.5-12.9%) in 2020-2021 in large rural towns/cities, 13.4% (12.3-14.6%) in 2015-2019 and 12.3% (10.5-14.5%) in 2020-2021 in small rural towns, and 9.7% (8.6-10.9%) in 2015-2019 and 10.9% (8.8-13.4%)in 2020-2021 isolated rural towns. The weighted prevalence of self-reported receipt of emergency food was 4.9% (4.8-5.1%) in 2015-2019 and 6.2% (5.8-6.5%) in 2020-2021 in urban areas, 6.8% (6.2-7.4%) in 2015-2019 and 7.6% (6.6-8.6%) in 2020-2021 in large rural towns/cities, 8.1% (7.3-9.1%) in 2015-2019 and 7.1% (5.7-8.8%) in 2020-2021 in small rural towns, and 6.8% (5.9-7.7%) in 2015-2019 and 8.5% (6.7-10.6%) in 2020-2021 isolated rural towns.

Conclusion: Households in rural communities use public and private food assistance at higher rates than urban areas, but there is variation across communities depending on the level of rurality.

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Introduction

In the United States, there are two major avenues for alleviating inadequate food access: public food assistance programs implemented by the government and the private emergency food system (sometimes referred to as public-private rather than private, as private organizations administer government food assistance programs). This working paper builds on our previous working paper (CES-25-01; Brady et al., 2025) describing food security across the rural-urban continuum pre- and mid-COVID-19 pandemic by describing the distribution of self-reported Supplemental Nutrition Assistance Program (SNAP) participation and receipt of emergency food by rurality/urbanicity.

SNAP is the largest federal food assistance program and served 12.5% of U.S. households in 2021 (Jones et al., 2022). SNAP has been shown to substantially reduce, but not eliminate, food insecurity (Gundersen et al., 2017a; Nord et al., 2009). However, there are numerous barriers to participation (Leung et al., 2013; Kaye et al., 2013). SNAP participation varies between urban and rural communities, with higher participation rates in rural areas (Food Research & Action Center, 2018; Vigil, 2019), which may partially be explained by socioeconomic differences along the rural-urban continuum. Notably, economic disruptions caused by the COVID-19 pandemic may have driven SNAP enrollment (Saloner et al, 2020). Further, enhanced benefits and administrative changes that expanded eligibility and reduced churn (when a household exits and re-enters SNAP) may also have contributed to increased enrollment and participation. All these factors could contribute the growth in SNAP participation since the pandemic began (Toossi et al, 2021; Jones & Toossi, 2024; USDA 2024). To our knowledge, there is no current literature on the distribution of changes in SNAP participation across the rural-urban continuum before and during the COVID-19 pandemic.

The emergency food system, also called the charitable food system, public-private food assistance system, or food banking system, provides another safety net for households with unmet food needs (Bazerghi et al., 2016; Fan et al., 2021). In 2020, 6.7% of households received emergency food from a food pantry (Coleman-Jensen, et al., 2021), but there are substantial barriers to accessing these resources (Bradley and Vitous, 2021; Fong et al., 2016; Kaplan et al., 2020; Neill et al., 2011; Wright et al., 2020).

Prior to the pandemic, use of emergency food resources also has been shown to be higher in more rural communities compared to urban areas (Gundersen et al., 2017b; Waxman et al., 2019). Among low-income households experiencing food insecurity there was a sharp overall increase in households receiving emergency food in 2020, but this was not observed for households in non-metropolitan communities (Brady et al., 2022).

We aimed to better understand the distribution of SNAP participation and use of emergency food resources across the rural-urban continuum before and during the pandemic. We answer the following two research questions: 1) What is the prevalence of households, including all households as well as low-income households only, who participated in the Supplemental Nutrition Assistance Program (SNAP) and who received emergency food at each level of the rural-urban continuum before and during the COVID-19 pandemic? And 2) What are the relationships between demographic factors, time period, and rural/urban status with SNAP participation and receipt of emergency food overall and for low-income households?

Data and Methodology

Data

The Current Population Survey (CPS) Food Security Supplement (FSS) is an annual survey examining food security and use of food assistance among households in the United States. It is a collaboration between the United States Census Bureau, Bureau of Labor Statistics, and United States Department of Agriculture. The survey collects data from approximately 30,000 households who are sampled each December. We link United States Department of Agriculture 2010 Rural Urban Commuting Area (RUCA) codes (available from https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes/) to households that completed the CPS FSS at the census tract level.

Sample

We pooled observations across CPS FSS years in a repeated cross-sectional design. We included households who completed the FFS for the first time between 2015 and 2021 in our analytic sample. We also examined a sub-sample of households earning under 185% of the Federal Poverty Threshold,

approximated with the midpoint of each household's income category and the number of individuals living in the households. There were about 1350,000 households in the analytic sample and about 38,500 households in the low-income sub-sample.

Measures

SNAP participation was a dichotomous variable with values of "Participant" and "Non-Participant" indicating self-reported participation in any month of the calendar year. Receipt of emergency food was based on two items: one asking about receipt of food from food pantries, food banks, or other community programs and the other asking about receipt of a meal from an emergency feeding program (such as a soup kitchen or meal site). Receipt of emergency food was a dichotomous variable with values of "Yes" or "No" indicating receipt of food from either source or not receiving emergency food from any sources in the previous year.

Time period was a dichotomous variable with values of "Pre-COVID" for survey years of 2015 to 2019 and "COVID" for survey years 2020 and 2021. Our coding for location is more fully described previously (Brady et al., 2025) and detailed in Table 2. Briefly, we used secondary RUCA codes that classify census tracts on the rural-urban continuum based on population size, urbanization, and primary and secondary daily commuting flows (USDA-ERS, n.d.) to define households as being in urban areas, large rural city/town, small rural town, or an isolated rural town/area.

We calculated adjusted prevalences using the following demographic characteristics: Food security status (food secure or food insecure) which was measured using the 18-item United States Department of Agriculture Household Food Security Survey Module (HFSSM) and classified based on their responses using USDA scoring methods (Bickel et al., 2000), household size (continuous), age of household respondent (34 and under, 35-59, 60 and older), sex of household respondent (Male or Female), racial self-classification of household respondent (White; Black; American Indian/Alaskan Native, Asian, Native Hawaiian and other Pacific Islander; or Other), Hispanic ethnicity of household respondent (Hispanic or non-Hispanic), education of household respondent (less than high school, high school or equivalent, some college or associates degree, or bachelor's degree or higher), household

income (Under $20,000, $20,000-$29,999, $30,000-$39,999, $40,000+), and employment status of household respondent (employed, unemployed, or not in labor force). We view the sex, racial self-classification, and ethnic self-classification of the household respondent variables as proxies for exposure to systemic and structural constraints on access to resources (Boyd et al., 2020), including food assistance resources, that influence food and economic security.

Analysis

First, we estimated the prevalence for SNAP participation or receipt of emergency food at each level of our urban/rural variable for each time period according to the following equation:

$$Weightedprevalence \in timeperiod\, j \wedge urban/rural\, codek = \frac{\sum_i y_{ijk} * w_{ijk}}{\sum_i w_{ijk}} \quad (1)$$

Where y_{ijk} is a binary measure of SNAP participation or receipt of emergency food for household *i*, in time period *j*, and in urban/rural code *k* and where w_{ijk} is the CPS FSS household weight for household *i*, in time period *j*, and in urban/rural code *k*. We then repeated this with our low-income sub-sample.

Then, we specified logistic regression models to evaluate SNAP participation or receipt of emergency food adjusting for key demographic characteristics. We applied CPS FSS household weights to account for survey design and non-response. Separate models were specified for the two outcomes of interest for the overall and low-income sub-sample as follows:

$$E(y\ \ ) = \alpha + \beta_1 t_j + \beta_2 r_k + \beta_3 (t_j * r_k) + \beta_4 State + \beta_5 f_i + \beta_6 X_i + \epsilon \quad (2)$$

Where y_{it} is the log odds of the outcome of interest (SNAP participation or receipt of emergency food) for household *I* in time *t*, t_j is the time period (2015-2019 or 2020-2021), r_k is the urban/rural code (urban areas, large rural city/town, small rural town, or an isolated rural town/area), State is a fixed effect for state of residence, f_i is the households' food security status, and $\boldsymbol{X}_i$ represents a vector of demographic characteristics for household *i* described above. The two-way interaction term between t_j and r_k is included to assess whether the outcomes varied by time period and urban/rural code. We examined odd

ratios and calculated predicted probabilities and 95% confidence intervals using the parameter estimates from the logistic regression models.

Results

Sample Demographics

Sample demographics are shown in Table 2. Nearly 85% of households lived in urban areas. Overall, approximately one in ten households participated in SNAP, while three in ten low-income households participated in SNAP. Receipt of emergency food was lower than participation in SNAP, with about 5% of all households receiving emergency food and 15% of low-income households receiving emergency food.

SNAP Participation

Estimated prevalences of households participating in SNAP for each rural classification in both time periods are shown in Table 3. Pre-COVID, SNAP participation was lower at the extreme ends of the rural-urban continuum (8.9% (95% CI: 8.7-9.2%) for urban areas, 9.7% (95% CI: 8.6-10.9%) for isolated rural towns/areas) and higher in the middle categories (11.4% (95% CI: 10.8-12.2%) for large rural towns/cities and 13.4% (95% CI: 12.3-14.6%) for small rural towns). In COVID, there were minimal changes in participation rates from Pre-COVID. Adjusted estimates in both time periods were similar across urban areas, large rural city/town, small rural town, and isolated rural town/area. For low-income households, unadjusted point estimates of SNAP participation were again highest in small rural towns. Isolated rural town/areas had lower participation compared to other areas Pre-COVID. During COVID, SNAP participation among low-income households was similar across rural-urban categories. Again, we observed similar estimates in the adjusted results. Regression tables used for obtained adjusted prevalences are included in Appendix 1.

Receipt of Emergency Food

Estimated prevalences of household receiving emergency food by time period for each rural classification are shown in Table 4. Overall, unadjusted estimates for receipt of emergency food ranged from 4.9% (95% CI: 4.8-5.1%) for urban areas to 8.1% (95% CI: 7.3-9.1%) for small rural towns Pre-

COVID. In COVID, they were again lowest in urban areas (6.2%, 95% CI:5.8-6.5%), but were highest in isolated rural towns/areas (8.5%, 6.7-10.6%). After adjustment, estimates were mostly similar across both rural-urban and time periods. Urban areas were the only classification where there was an increase in the adjusted prevalence of households receiving emergency food between time periods. For low-income households, unadjusted estimates for receipt of emergency food were also lowest in urban areas (12.2%, 95% CI: 12.7-13.8%) and highest for small rural towns (17.0%, 95% CI: 15.1-19.0%) Pre-COVID. During COVID, the only apparent increase was in the prevalence was for urban areas (16.8%, 95% CI: 15.8-17.9%). Again, we observed similar estimates in the adjusted estimates. Regression tables used for obtained adjusted prevalences are included in Appendix 1.

Conclusions and Future Work

We estimated self-reported receipt of public and private food assistance across the rural-urban continuum before and during the COVID-19 pandemic. Substantial economic supports for households and community-based emergency food providers were implemented in response to concerns that food insecurity would increase during the early days of the COVID-19 pandemic (Niles et al, 2020). We show that the prevalence of SNAP participation and receipt of food from emergency food providers were relatively stable comparing pre-COVID and during the COVID period. Additionally, participation was generally higher in rural towns/cities. Our findings are complicated by the overall improvements to the economic social safety net during this time but offer insight into use of different food assistance resources during the pandemic in urban and rural communities.

SNAP participation was higher in large rural towns/cities and small rural towns than urban or isolated rural areas both before and during the pandemic. The difference between isolated rural areas and small and large rural towns is a key finding of this analysis, though the reason for these differences remains unclear. For example, it is possible that households living in isolated rural areas may have greater difficulty accessing SNAP benefits and authorized retailers compared to households living in even modestly more densely populated areas, like rural towns. Alternatively, there may be key differences in economic security based on degree of rurality. Overall there is limited research looking at rural

communities; more research on the factors that contribute to rural heterogeneity in use of food assistnace, including SNAP, is warranted. We also show that self-reported SNAP participation did not meaningfully change between 2015 to 2019 and 2020 and 2021 at any level of the rural-urban continuum. This reflects other research analyzing primary survey data collected before and during the pandemic that showed similar rates of self-reported SNAP participation (Ohri-Vachaspati et al, 2021) . In contrast, analyses of administrative data showed relatively small but meaningful increases in participation throughout the pandemic (Jones & Toossi, 2024). Misreporting in self-reported survey data could explain this discrepancy. Further research could explore the complexities of these issues by examining differences in urban and rural communities using administrative data or linked self-reported survey and administrative data extending beyond 2021.

Self-reported receipt of emergency food was lowest in urban areas prior to the pandemic, but increased to more closely resemble estimates from rural communities in 2020 and 2021. This is in line with previous research that observed an overall increase in receipt of emergency food in 2020 compared to previous years, but not for non-metropolitan households (Brady et al, 2022). The availability of alternate methods for the distribution of emergency food resources (e.g., drive thru mass distribution events) may have facilitated increased access to this resource for residents in urban areas more so than in rural communities. Future work should evaluate the link between the availability and type of emergency food providers in different communities, and what impact this had on receipt of emergency food.

Overall, these results should be viewed with several limitations in mind. First, we relied on self-reported SNAP participation and receipt of emergency food. This likely underestimated the true prevalence of households accessing these resources, but it is unclear whether this would meaningfully differ based on urbanicity or rurality. Additionally, while our measure of the urban-rural continuum was more detailed than dichotomizing into metropolitan and non-metropolitan households, aggregating households still masks important differences between communities. While beyond the scope of this work, more detailed classification based on community characteristics such as sociodemographic profiles would complement these findings.

Overall, we found less variation across types of communities in the prevalences of households reporting that they participated in SNAP or received emergency food during the pandemic. In all communities, it appears that a substantial portion of low-income households who may benefit from food assistance, whether public or private, are not utilizing these resources. At the same time, increasing use of these resources will have costs and may strain already taxed social service providers. Developing sustainable, place-based solutions that fit community contexts will be required to support households' food security and connect underserved households to public and private food assistance resources.

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Table 1. Rural classification based on Secondary Rural-Urban Area Commuting Codes

Rural Classification	Description	Rural-Urban Area Commuting Codes
Urban	Households living in metropolitan area or households living in micropolitan areas, small rural towns, or isolated rural towns/areas with substantial commuting flows to metropolitan areas	1.0, 1.1, 2.0, 2.1, 3.0, 4.1, 5.1, 7.1, 8.1, 10.1
Large rural city/town	Households living in micropolitan area or households in small rural towns or isolated rural towns/areas with substantial commuting flows to micropolitan areas	4.0, 5.0, 6.0, 7.2, 8.2, 10.2
Small rural city/town	Households living in small rural towns or households in isolated rural towns/areas with substantial commuting flows to small rural towns	7.0, 8.0, 9.0, 10.3
Isolated rural town/area	Households living in isolated rural towns/areas without substantial commuting flows to metropolitan areas, micropolitan areas, or small rural towns.	10.0

Table 2. Weighted descriptive statistics for households overall and for households with incomes less that approximately 185% of the Federal Poverty Threshold from pooled data from the Current Population Survey Food Security Supplement from 2015 to 2021

Characteristic	All households	Low-income households
Rural status, %		
Urban	84.7	80.5
Large rural city/town	8.8	10.8
Small rural city/town	3.7	5.2
Isolated rural town/area	2.8	3.5
Time period, %		
2015-2019	71.5	72.8
2020 & 2021	28.5	27.2
Food Security Status, %		
Food Secure		
Food Insecure		
Age category, %		
Under 35	23.3	28.5
35-59	43.1	37.3
60 and older	33.6	34.2
Gender, %		
Male	49.2	41.4
Female	50.8	58.6
Race, %		
White	67.6	61.2
Black	12.0	17.7
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	5.9	6.0
Multiracial	14.5	155.1
Hispanic ethnicity, %		
Yes	14.2	22.5
No	85.8	77.5
Yearly income, %		
Less than $5000	3.0	10.3
$5000-$7499	1.5	4.9
$7500-$9999	2.3	7.7
$10,000-$12,499	2.9	10.0
$12,500-$14,999	2.7	9.0
$15,000-$19,999	4.2	14.2
$20,000-$24,999	5.3	13.7
$25,000-$29,999	5.3	11.1
$30,000-$34,999	5.7	6.4
$35,000-$39,999	5.4	5.4
$40,000-$49,999	8.1	4.8
$50,000 or more	53.6	2.5
Employment status, %		
Employed	61.5	42.2
Unemployed	2.9	5.1
Not in Labor force	35.6	52.7
Household size, mean	2.48	2.70

Table 3. Weighted prevalence of households participating in the Supplemental Nutrition Assistance Program (SNAP) pre- and during the COVID-19 pandemic

Rural Classification*	Prevalence of Households Participating in SNAP		Adjusted[1] Prevalence of Households Participating in SNAP	
	2015-2019	2020-2021	2015-2019	2020-2021
All households				
Urban	8.9 [8.7-9.2]	9.1 [8.7-9.5]	9.2 [9.0-9.4]	9.3 [8.9-9.6]
Large Rural Town/City	11.4 [10.8-12.2]	11.6 [10.5-12.9]	10.0 [9.4-10.5]	10.2 [9.3-11.2]
Small Rural Town	13.4 [12.3-14.6]	12.3 [10.5-14.5]	10.4 [9.7-11.2]	10.2 [8.8-11.7]
Isolated Rural Town/Area	9.7 [8.6-10.9]	10.9 [8.8-13.4]	9.1 [8.2-10.0]	10.9 [8.9-12.8]
Low-income households				
Urban	27.1 [26.4-27.8]	28.1 [26.9-29.4]	26.8 [26.2-27.4]	27.7 [26.6-28.9]
Large Rural Town/City	27.5 [25.9-29.1]	29.3 [26.4-32.4]	28.6 [27.0-30.1]	30.4 [27.6-33.3]
Small Rural Town	29.5 [27.2-32.0]	28.2 [23.9-32.8]	30.0 [27.8-32.1]	29.8 [25.6-34.1]
Isolated Rural Town/Area	23.6 [21.1-26.2]	26.2 [21.2-32.0]	25.9 [23.4-28.5]	30.2]24.6-35.7]

*Rural classification is based on secondary United States Department of Agriculture Rural Urban Commuting Area codes, which consider urbanicity, population size, and primary and secondary daily commuting flows

1.Adjusted prevalences were obtained from marginal effects based on regression models predicting food security status as a function of rural status, time period, their interaction, and adjusting for state of residence, household size, age of household respondents, sex of household respondent, racial self-classification of household respondent, Hispanic ethnicity of household respondent, household income, and employment status of household respondents.

Table 4. Weighted prevalence of households receiving emergency foody pre- and during the COVID-19 pandemic

Rural Classification*	Prevalence of Households Receiving Emergency Food		Adjusted[1] Prevalence of Households Receiving Emergency Food	
	2015-2019	2020-2021	2015-2019	2020-2021
All households				
Urban	4.9 [4.8-5.1]	6.2 [5.8-6.5]	5.1 [4.9-5.3]	6.3 [6.0-6.7]
Large Rural Town/City	6.8 [6.2-7.4]	7.6 [6.6-8.6]	5.9 [5.4-6.3]	6.5 [5.7-7.3]
Small Rural Town	8.1 [7.3-9.1]	7.1 [5.7-8.8]	6.0 [5.4-6.7]	5.9 [4.8-7.0]
Isolated Rural Town/Area	6.8 [5.9-7.7]	8.5 [6.7-10.6]	6.0 [5.3-6.7]	8.4 [6.6-10.2]
Low-income households				
Urban	12.2 [12.7-13.8]	16.8 [15.8-17.9]	13.3 [12.8-13.7]	16.7 [15.7-17.6]
Large Rural Town/City	14.9 [13.7-16.3]	17.1 [14.8-19.7]	15.2 [14.0-16.5]	17.3 [14.9-19.7]
Small Rural Town	17.0 [15.1-19.0]	15.6 [12.4-19.5]	15.8 [14.1-17.5]	16.0 [12.8-19.3]
Isolated Rural Town/Area	15.3 [13.3-17.6]	17.5 [13.5-22.3]	15.3 [13.3-17.3]	19.4 [14.8-24.1]

*Rural classification is based on secondary United States Department of Agriculture Rural Urban Commuting Area codes, which consider urbanicity, population size, and primary and secondary daily commuting flows

1.Adjusted prevalences were obtained from marginal effects based on regression models predicting food security status as a function of rural status, time period, their interaction, and adjusting for state of residence, household size, age of household respondents, sex of household respondent, racial self-classification of household respondent, Hispanic ethnicity of household respondent, household income, and employment status of household respondents.

Appendix 1

Table A1. Regression results from weighted models predicting SNAP participation as a function of rural status, time period, their interaction, and adjusting for state of residence and demographic characteristics based on rural classification using urbanicity, population size, and primary daily commuting flows

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	1.143	0.056	1.037	1.259
Small Rural Town	1.229	0.080	1.081	1.397
Isolated Rural Town/Area	0.987	0.082	0.838	1.161
Time Period (Ref=2015-2019)				
2020-2021	1.272	0.049	1.181	1.371
Rural Classification x Time Period				
Large Rural Town x 2020-2021	1.030	0.099	0.853	1.243
Small Rural Town x 2020-2021	0.953	0.133	0.725	1.252
Isolated Rural Town/Area x 2020-2021	1.317	0.223	0.935	1.853
Food Security Status (ref=Food insecure)				
Food secure	0.200	0.006	0.0188	0.212
Age (Ref=34 and under)				
35 to 59	1.044	0.039	0.970	1.123
60 and older	0.636	0.028	0.583	0.693
Sex (Ref=Male)				
Female	1.561	0.046	1.473	1.653
Racial self-classification (Ref=white)				
Black	2.022	0.086	1.861	2.198
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	0.775	0.057	0.671	0.895
Other	1.279	0.049	1.189	1.377
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	1.158	0.049	1.066	1.258
Income category (Ref=less than $5,000)				
$5000-$7499	1.095	0.093	0.928	1.294
$7500-$9999	1.879	0.138	1.627	2.170
$10,000-$12,499	1.458	0.099	1.277	1.665
$12,500-$14,999	1.037	0.074	0.903	1.192
$15,000-$19,999	0.713	0.046	0.626	0.813
$20,000-$24,999	0.552	0.036	0.487	0.627
$25,000-$29,999	0.363	0.025	0.317	0.415
$30,000-$34,999	0.258	0.018	0.225	0.297
$35,000-$39,999	0.191	0.015	0.165	0.222
$40,000-$49,999	0.135	0.010	0.117	0.156
$50,000 or more	0.041	0.003	0.035	0.046
Employment status (Ref=employed)				
Unemployed	2.464	0.166	2.160	2.812
Not in the labor force	1.898	0.066	1.774	2.031
Household size	1.550	0.016	1.519	1.581

Notes: Model also included fixed effects for state of residence

Appendix 1

Table A2. Regression results from weighted models predicting SNAP participation as a function of rural status, time period, their interaction, and adjusting for state of residence and demographic characteristics based on rural classification using urbanicity, population size, and primary daily commuting flows among households incomes approximately less than 185% of the Federal Poverty Threshold

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	1.118	0.058	1.009	1.238
Small Rural Town	1.216	0.083	1.063	1.391
Isolated Rural Town/Area	0.947	0.081	0.801	1.120
Time Period (Ref=2015-2019)				
2020-2021	1.194	0.050	1.010	1.297
Rural Classification x Time Period				
Large Rural Town x 2020-2021	1.053	0.110	0.859	1.291
Small Rural Town x 2020-2021	0.933	0.139	0.697	1.249
Isolated Rural Town/Area x 2020-2021	1.224	0.231	0.847	1.771
Food Security Status (ref=Food insecure)				
Food secure	0.259	0.008	0.243	0.276
Age (Ref=34 and under)				
35 to 59	1.132	0.044	1.049	1.223
60 and older	0.684	0.032	0.625	0.749
Sex (Ref=Male)				
Female	1.570	0.050	1.475	1.671
Racial self-classification (Ref=white)				
Black	1.914	0.088	1.750	2.093
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	0.756	0.068	0.652	0.876
Other	1.316	0.055	1.214	1.428
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	1.119	0.050	1.026	1.221
Income category (Ref=less than $5,000)				
$5000-$7499	1.095	0.086	0.938	1.278
$7500-$9999	1.767	0.121	1.544	2.021
$10,000-$12,499	1.392	0.088	1.229	1.576
$12,500-$14,999	0.996	0.066	0.875	1.135
$15,000-$19,999	0.710	0.044	0.628	0.802
$20,000-$24,999	0.629	0.040	0.556	0.712
$25,000-$29,999	0.471	0.033	0.411	0.539
$30,000-$34,999	0.392	0.032	0.334	0.461
$35,000-$39,999	0.310	0.028	0.260	0.371
$40,000-$49,999	0.262	0.025	0.217	0.316
$50,000 or more	0.158	0.022	0.121	0.208
Employment status (Ref=employed)				
Unemployed	2.150	0.150	1.876	2.465
Not in the labor force	1.778	0.066	1.654	1.912
Household size	1.388	0.017	1.355	1.423

Notes: Model also included fixed effects for state of residence

Appendix 1

Table A3. Regression results from weighted models predicting receipt of emergency food as a function of rural status, time period, their interaction, and adjusting for state of residence and demographic characteristics based on rural classification using urbanicity, population size, and primary daily commuting flows

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	1.212	0.072	1.079	1.361
Small Rural Town	1.258	0.099	1.078	1.467
Isolated Rural Town/Area	1.242	0.111	1.043	1.479
Time Period (Ref=2015-2019)				
2020-2021	1.591	0.070	1.461	1.734
Rural Classification x Time Period				
Large Rural Town x 2020-2021	0.857	0.096	0.688	1.067
Small Rural Town x 2020-2021	0.720	0.113	0.529	0.980
Isolated Rural Town/Area x 2020-2021	1.210	0.228	0.836	1.751
Food Security Status (ref=Food insecure)				
Food secure	0.110	0.004	0.102	0.118
Age (Ref=34 and under)				
35 to 59	1.579	0.073	1.443	1.728
60 and older	0.926	0.052	0.831	1.033
Sex (Ref=Male)				
Female	1.164	0.040	1.088	1.246
Racial self-classification (Ref=white)				
Black	1.857	0.094	1.682	2.051
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	0.813	0.071	0.685	0.964
Other	1.173	0.054	1.072	1.284
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	1.199	0.061	1.085	1.325
Income category (Ref=less than $5,000)				
$5000-$7499	0.880	0.091	0.719	1.078
$7500-$9999	1.031	0.089	0.871	1.220
$10,000-$12,499	0.976	0.080	0.931	1.145
$12,500-$14,999	0.809	0.071	0.681	0.960
$15,000-$19,999	0.712	0.056	0.610	0.831
$20,000-$24,999	0.624	0.049	0.534	0.728
$25,000-$29,999	0.509	0.042	0.433	0.598
$30,000-$34,999	0.440	0.037	0.373	0.519
$35,000-$39,999	0.306	0.028	0.256	0.366
$40,000-$49,999	0.275	0.024	0.232	0.325
$50,000 or more	0.118	0.009	0.101	0.138
Employment status (Ref=employed)				
Unemployed	1.853	0.139	1.599	2.147
Not in the labor force	1.763	0.074	1.623	1.915
Household size	1.166	0.139	1.139	1.194

Notes: Model also included fixed effects for state of residence

Appendix 1

Table A4. Regression results from weighted models predicting receipt of emergency food as a function of rural status, time period, their interaction, and adjusting for state of residence and demographic characteristics based on rural classification using urbanicity, population size, and primary daily commuting flows among households incomes approximately less than 185% of the Federal Poverty Threshold

Variable	Odds Ratio	STD ERR	95% CI LB	95% CI UB
Rural Classification (Ref=Urban)				
Large Rural Town/City	1.216	0.079	1.070	1.382
Small Rural Town	1.281	0.109	1.084	1.514
Isolated Rural Town/Area	1.222	0.120	1.008	1.483
Time Period (Ref=2015-2019)				
2020-2021	1.500	0.076	1.359	1.657
Rural Classification x Time Period	0.870	0.111	0.677	1.117
Large Rural Town x 2020-2021	0.738	0.130	0.523	1.041
Small Rural Town x 2020-2021	1.028	0.217	0.679	1.556
Isolated Rural Town/Area x 2020-2021				
Food Security Status (ref=Food insecure)				
Food secure	0.154	0.006	0.143	0.166
Age (Ref=34 and under)				
35 to 59	1.740	0.087	1.578	1.920
60 and older	1.020	0.061	0.907	1.148
Sex (Ref=Male)				
Female	1.133	0.045	1.050	1.225
Racial self-classification (Ref=white)				
Black	1.622	0.091	1.453	1.809
Asian, American Indian/Alaska Native, Native Hawaiian/Pacific Islander	0.735	0.070	0.610	0.886
Other	1.176	0.062	1.062	1.304
Hispanic ethnicity (Ref=non-Hispanic)				
Hispanic	1.086	0.060	0.974	1.210
Income category (Ref=less than $5,000)				
$5000-$7499	0.896	0.087	0.741	1.082
$7500-$9999	1.016	0.082	0.867	1.190
$10,000-$12,499	0.951	0.073	0.818	1.106
$12,500-$14,999	0.782	0.064	0.666	0.919
$15,000-$19,999	0.699	0.052	0.604	0.809
$20,000-$24,999	0.646	0.051	0.553	0.754
$25,000-$29,999	0.549	0.047	0.465	0.649
$30,000-$34,999	0.578	0.059	0.473	0.705
$35,000-$39,999	0.411	0.048	0.328	0.516
$40,000-$49,999	0.435	0.052	0.343	0.551
$50,000 or more	0.383	0.064	0.277	0.530
Employment status (Ref=employed)				
Unemployed	1.795	0.144	1.533	2.101
Not in the labor force	1.640	0.077	1.496	1.798
Household size	1.097	0.016	1.066	1.128

Notes: Model also included fixed effects for state of residence